Artificial Intelligence (AI-2002)

Lecture 1: Introduction to Artificial Intelligence

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Class Rules:

- Attendance will not be accepted if you arrive more than 10 minutes after the class begins.
- Retaking quizzes and assignments is not permitted.
- If time allows, there will be an activity at the end of the class.

Marks Distribution:

THEORY

Assessment	Marks
Finals	50
Mids	30
Quizzes 5 total – 4 best	8
Assignments	8
Hackathon/ Project	4

Text Book:

- Stuart Russell and Peter Norvig, Artificial Intelligence. A Modern Approach, 4th Global Edition, Prentice Hall, Inc.

What is Artificial Intelligence?

What is Artificial Intelligence?

- Humans, known as Homo sapiens ("wise man"), are defined by intelligence, driven to understand how our complex brain enables thought, learning, and interaction.
- AI is the study and creation of intelligent systems that replicate or enhance human intelligence to think, understand, and act effectively in various situations.
- AI is a rapidly advancing field with a trillion-dollar annual impact, and visionaries like Kai-Fu Lee believe its influence will surpass all previous technologies in history.

What is Artificial Intelligence?

- Unlike established sciences, AI is still in its early stages, offering abundant opportunities for new ideas and innovations, making it exciting for students and researchers.
- AI encompasses general areas like learning, reasoning, and perception, as well as specific tasks such as chess, math, poetry, autonomous driving, and medical diagnosis.

What is Artificial Intelligence?

- AI is a field that explores how machines can exhibit intelligent behavior. However, defining what constitutes intelligence is a subject of debate. Some researchers define intelligence based on how closely it mimics human performance, while others prefer a more abstract definition, which is based on rationality—doing the "right thing" or making optimal decisions.

Human vs. Rational Intelligence

- **Human Intelligence:** This focuses on replicating human-like thought processes and behavior.
- **Rational Intelligence:** This focuses on the idea of making optimal, logical decisions, regardless of whether the process mimics human reasoning.

Thought vs. Behavior

Q: Is a machine intelligent because it thinks intelligently or behaves intelligently?

Thought vs. Behavior

- **Thought:** This focuses on the internal cognitive processes involved in reasoning, decision-making, and problem-solving.
- **Behavior:** This focuses on observable actions and behaviors that demonstrate intelligent activity, regardless of the underlying thought process.

- By combining the two distinctions (human vs. rational and thought vs. behavior), there are four possible combinations, each with its own focus:
- **Human + Thought:** Studying and replicating human thought processes.
- **Human + Behavior:** Observing and replicating human actions and behaviors.
- **Rational + Thought:** Focusing on creating rational decision-making systems that may not mimic human thinking.
- **Rational + Behavior:** Focusing on rational actions, or doing the right thing, with minimal focus on how the system thinks.

Acting humanly: The Turing test approach

- Proposed by Alan Turing in 1950, the Turing Test is a thought experiment aimed at answering the question, "Can a machine think?" instead of delving into the philosophical complexities.
- The test involves a human interrogator who asks questions to both a human and a machine, and then tries to determine which is which based on their written responses. If the interrogator cannot reliably tell, the machine is said to have passed the test.

Acting humanly: The Turing test approach

- The Turing Test is significant because it focuses on behavior—whether a machine can imitate human-like intelligence through conversation—rather than attempting to simulate the internal processes of human thought.

Capabilities Required for Passing the Turing Test:

- Natural Language Processing (NLP): This allows the machine to understand and communicate effectively in human language.
- Knowledge Representation: The machine needs a way to store and recall information it knows or hears in a structured manner.
- Automated Reasoning: The machine should be able to answer questions and draw conclusions based on the knowledge it has.
- Machine Learning: The machine must be able to adapt to new circumstances and detect patterns in data, allowing it to improve over time.

- While the Turing Test is an important benchmark, AI researchers have focused more on understanding the underlying principles of intelligence rather than simply trying to pass the test. This is similar to the history of aeronautical engineering, where progress came not from trying to make airplanes that mimic birds exactly but from understanding the science of aerodynamics and using tools like wind tunnels.
- In the same way, AI researchers often prioritize the fundamental principles of intelligence, such as learning, reasoning, and perception, rather than merely focusing on whether a machine can "fool" a human into thinking it is a person.

Thinking humanly: The cognitive modeling approach

To create a program that thinks like a human, we first need to understand how human thinking works. This can be studied through three main methods:

- **Introspection:** Observing and analyzing your own thought processes as they happen.
- **Psychological Experiments:** Studying how people behave and solve problems through controlled experiments.
- **Brain Imaging:** Using technologies like fMRI or EEG to observe brain activity in real-time as people think and act.

Thinking humanly: The cognitive modeling approach

- Once a theory of human thought is formulated, it can be translated into a computer program. If the program behaves like a human (in terms of both output and the process used to reach that output), this suggests that the program might mimic some mechanisms of human cognition.
- Cognitive science aims to create precise and testable theories of the human mind. It borrows techniques from neuroscience, psychology, and AI to study cognition comprehensively.

Thinking humanly: The cognitive modeling approach

Early AI researchers sometimes confused good performance on tasks with accurate models of human thinking. Modern research distinguishes between:

- **AI:** Focused on creating systems that perform tasks effectively, whether or not they mimic human cognition.
- **Cognitive Science:** Dedicated to studying and modeling how humans actually think.

Thinking rationally

- The thinking rationally focuses on creating AI systems that can think correctly and make logical decisions based on given information. Although perfect knowledge is rare, AI uses tools like probability to reason with uncertainty. However, reasoning alone isn’t enough; intelligent systems also need strategies to act effectively in real-world scenarios. This approach is one of the foundational ideas in the development of AI.

Acting rationally

- A rational agent is one that acts in a way to achieve the best possible outcome or, if uncertainty exists, the best expected outcome.

Basic Components of AI

- **Perception:** The ability to take in and interpret data from the environment (e.g., vision, sound, or sensors).
- **Learning:** The process of improving performance based on data and experience. Machine learning is a major part of this, where the system learns patterns from data and adapts over time.
- **Reasoning:** The ability to make decisions or draw conclusions from available information. This involves logic and problem-solving, like a chess AI determining the best move based on the current board.

Basic Components of AI

- **Knowledge Representation:** How the AI stores information in a structured way, such as facts, rules, or models of the world. For instance, a medical AI might store knowledge about diseases and symptoms in a database.
- **Action:** The ability to take actions or perform tasks based on reasoning and learning. For example, a robot performing tasks like assembling products or a chatbot responding to a user's query.

Types of Artificial Intelligence

- **Narrow AI (Weak AI):** This type of AI is designed and trained for a specific task. It operates within a limited domain and cannot perform tasks outside of its programming. Examples include virtual assistants (like Siri, Alexa) and recommendation systems.
- **General AI (Strong AI):** This type of AI has the ability to understand, learn, and apply intelligence across a wide range of tasks, similar to human cognition. It does not currently exist but is a goal for future AI research.

Types of Artificial Intelligence

- **Superintelligent AI:** This refers to an AI that surpasses human intelligence in every field, including creativity, problem-solving, and decision-making. It is a theoretical concept and is not yet achieved.

Q: Think about an AI application and identify whether it replicates human behavior or follows rationality.

Next:

- Quiz: 1